

Introduction to B-Tree & B+ Tree Data Structures

> **Definition:** A **B-Tree** is a self-balancing multi-way search tree of order m where each node can contain multiple keys and more than two children, designed specifically to optimize **disk read/write block accesses** in database indexing and file systems.

1. Detailed Technical Explanation

1. Properties of a B-Tree of Order m :

1. Every node has at most m children.
2. Every other node, except the root and leaf nodes, must have at least $\lceil m/2 \rceil$ children.

2. B-Tree vs B+ Tree Comparison

B+ TREE:

Detailed Comparison Table:

Feature	B-Tree	B+ Tree
Search Path	Search path may vary (depends on leaf location)	Search path is uniform (all leaf nodes at same level)

3. Core Concepts & Memory Keywords

- **Order m:** Maximum number of children a node can have.

- **F**

4. Must-Write Points for Exams

- B-Trees and B+ Trees reduce disk I/O by fitting large numbers of keys inside a single disk page block (e.g., 4KB or 8KB).

- B+

5. Quick Recall Flow

Multi-V